

ISMAIL ELEZI

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I am a Principal Research Scientist of Computer Vision, leading the multi-modality learning team in Huawei Noah's Ark Lab in London. I worked before in deep metric learning, self- and semi-supervised learning, open-world and long-tail detection, active learning, and generative image models. Currently I am focused on researching new models in multi-modality learning (visual LLMs). I frequently publish in top-tier vision (CVPR, ICCV, ECCV) and ML conferences (NeurIPS, ICML, ICLR, AAAI). ***h-index: 18, citations: 1601.***

WORKING EXPERIENCE

Deep Learning Principal Researcher, Huawei

Dec 2024 – Present

IC/manager hybrid role where I am leading a team of research scientists and research interns working in multi-modality learning, VLMs and diffusion models. Main target of the team is to build efficient solutions for (V)LLMs, and we target our work at top-tier conferences. Currently very interested in diffusion LLMs.

London, UK

Deep Learning Senior Researcher, Huawei

Apr 2023 – Nov 2024

Started as an IC working on 3D computer vision. Promoted to team-lead within 9 months working on LLMs, tinyML and long-tail learning. Was tasked with building a multi-modality learning team.

London, UK

Deep Learning Research Visitor, Argo AI

Jun 2022 – Sep 2022

Worked on semi-supervised LiDAR data for object segmentation. Build a 3D masked-autoencoder solution for segmentation based on LiDAR data.

Munich, Germany

Alexander von Humboldt Postdoctoral Researcher, TUM

Oct 2020 – Mar 2023

Had my own projects in active learning and metric learning, while also mentored and collaborated with 3 PhD students and a dozen master students. Co-taught 3 different courses, and helped my supervisor in writing the ERC grant.

Munich, Germany

Deep Learning Research Intern, NVIDIA

Feb 2020 – Sep 2020

Worked on active learning and semi-supervised learning, resulting in 2 top-tier papers. Furthermore, helped on the technology transfer within the company.

Santa Clara, USA

EDUCATION

PhD in Deep Learning

Sep 2016 – Jul 2020

Ca' Foscari University of Venice, Venice, Italy

Distinction

- Completed PhD under the supervision of Marcello Pelillo and Thilo Stadelmann, working mostly on semi-supervised learning and metric learning. Spent a year as research visitor at TUM, and interned at Nvidia.

SELECTED PUBLICATIONS

1. Miles, Toker, Oncescu, Deng, and **Elezi**. Test-time scaling with diffusion language models via reward-guided stitching. *arXiv:2602.22871*, 2026
2. Gaintseva, Stepanov, Liu, Benning, Slabaugh, Deng, and **Elezi**. Midsteer: Optimal affine framework for steering generative models. *ICML26*
3. Choi, Miles, Potamias, **Elezi**, Deng, and Zafeiriou. Do you see what i am pointing at? gesture-based egocentric video question answering. *CVPR26*
4. Gaintseva, Oncescu, Ma, Liu, Benning, Slabaugh, Deng, and **Elezi**. Casteer: Steering diffusion models for controllable generation. *ICLR26*
5. Paul, Murphy, Gritta, Cardenas, Prokhorov, Bolliger, Toker, Miles, Oncescu, Sivakumar, Borchert, **Elezi**, Zhang, Lee, Zhang, Lampouras, and Wang. A benchmark for deep information synthesis. *ICLR26*
6. Xie, Yang, Yang, Deng, Dai, Gu, Wang, An, Zhao, Fend, Miles, **Elezi**, and Deng. Victor: Improving visual comprehension via token reconstruction for pretraining lmms. *AAAI26*
7. Xie, Yang, An, Wu, Zhao, Deng, Ran, Wang, Feng, Miles, **Elezi**, and Deng. Region-based cluster discrimination for visual representation learning. *ICCV25*

8. Wen, Zhao, **Elezi**, Deng, and Qi. Principal components enable a new language of images. *ICCV25*
9. Alexandridis, **Elezi**, Deng, Nguyen, and Luo. Fractal calibration for long-tailed object detection. *CVPR25*
10. Kaul, Ma, **Elezi**, and Deng. From attention to activation: Unravelling the enigmas of large language models. *ICLR25*
11. Miles, Reddy, **Elezi**, and Deng. Velora: Memory efficient training using rank-1 sub-token projections. *NeurIPS24*
12. Reddy*, **Elezi***, and Deng. G3dr: Generative 3d reconstruction in imagenet. *CVPR24*
13. Miles, **Elezi**, and Deng. Vkd: Improving knowledge distillation using orthogonal projections. *CVPR24*
14. Ma, **Elezi**, Deng, Dong, and Xu. Three heads are better than one: Complementary experts for long-tailed semi-supervised learning. *AAAI24*
15. Seidenschwarz, Brasó, Serrano, **Elezi**, and Leal-Taixé. Simple cues lead to a strong multi-object tracker. *CVPR23*
16. **Elezi**, Seidenschwarz, Wagner, Vascon, Torcinovich, Pelillo, and Leal-Taixé. The group loss++: A deeper look into group loss for deep metric learning. *tPAMI23*
17. Kocsis, Sukenik, Brasó, Niessner, Leal-Taixé, and **Elezi**. The unreasonable effectiveness of fully-connected layers for low-data regimes. *NeurIPS22*
18. Fomenko, **Elezi**, Ramanan, Osep, and Leal-Taixé. Learning to discover and detect objects. *NeurIPS22*
19. **Elezi**, Yu, Anandkumar, Leal-Taixé, and Alvarez. Not all labels are equal: Rationalizing the labeling costs for training object detection. *CVPR22*
20. Choi, **Elezi**, Lee, Farabet, and Alvarez. Active learning for deep object detection via probabilistic modeling. *ICCV21*
21. Seidenschwarz, **Elezi**, and Leal-Taixé. Learning intra-batch connections for deep metric learning. *ICML21*
22. **Elezi**, Vascon, Torcinovich, Pelillo, and Leal-Taixé. The group loss for deep metric learning. *ECCV20*
23. Maximov*, **Elezi***, and Leal-Taixé. CIAGAN: conditional identity anonymization generative adversarial networks. *CVPR20*

SKILLS

Programming: Python, PyTorch, TensorFlow, OpenCV, Scikit-learn, Java, Matlab/Octave, C, C#, C++.

Computer: Linux (Debian), Docker, AWS.

REVIEWING DUTIES

Area Chair (Conferences): WACV 2021; NeurIPS 2025, 2026

Reviewer (Conferences): CVPR 2020 - ; ICCV 2021 - ; ECCV 2022 - ; NeurIPS 2021 - ; ICML 2022 - ; ICLR 2024 -

SELECTED INTERNS AND THESISISTS SUPERVISED

2025: Ye Mao (Imperial College London); Mohammad Sadil Khan (University of Kaiserslautern); Yura Choi (Imperial College London); Ye-Bin Moon (POSTECH); Xin Wen (University of Hong Kong); Tatiana Gaintseva (Queen Mary University).

2024: Aysim Toker (Technical University of Munich → Huawei); Changrui Chen (University of Warwick → Huawei); Bingchen Zhao (University of Edinburgh); Prannay Kaul (2024) (University of Oxford → Amazon); Yongshuo Zong (University of Edinburgh).

2023: Konstantinos Alexandridis (King's College → Huawei); Roy Miles (Imperial College London → Huawei); Chengcheng Ma (CAS → Alibaba); Yunqi Miao (University of Warwick → Huawei).

PhD TUM 2020-2022: Jenny Seidenschwarz → Odyssey; Franziska Gerken.

Master TUM 2020-2022: Volodymyr Fomenko → OpenAI; Peter Kocsis → PhD at TUM; Laurin Wagner → myReha; Peter Sukenik → IST Austria; Feliks Hibraj → Snap Inc.